Mechanical Design Engineer Study Guide (All-in-One)

1. Core Fundamentals

- Strength of Materials (SOM): Stress, strain, bending, torsion, fatigue formulas.
- Theory of Machines (TOM): Kinematics, dynamics, vibrations.
- Machine Design: Shafts, bearings, gears, fasteners, springs, welds.
- Thermal & Heat Transfer essentials.
- Material science: properties, failure modes, selection.

2. Design Data & Handbooks

- Important extracts from PSG Design Data Book / Machinery's Handbook.
- Standard mechanical properties of materials.
- Safety factors and load calculation methods.

3. CAD / CAE

- CAD: SolidWorks, CATIA, Creo, NX basics.
- CAE: FEA (stress, thermal, modal, fatigue analysis).
- GD&T;: Geometric Dimensioning & Tolerancing (ASME Y14.5 overview).

4. Manufacturing & DFM/DFA

- Processes: Casting, machining, welding, additive manufacturing.
- Design for Manufacturing (DFM) & Design for Assembly (DFA) checklist.

5. Industry Standards & Documentation

- ISO / ASME standards for design.
- Bill of Materials (BOM) format and usage.
- Drawing standards and title block examples.

6. Practical Case Studies & Examples

- Example: Designing a shaft with safety factor.
- Example: Vibration analysis of two-wheeler mudguard.
- Example: Stress calculation for a bracket.

7. Study Roadmap (3-Month Plan)

Month 1: Strength of Materials, Machine Design basics, CAD modeling practice (daily small parts).
Month 2: FEA basics, vibration & thermal analysis, DFM/DFA study, manufacturing processes.
Month 3: Advanced CAD assemblies, GD&T; practice, mini-projects (case studies), portfolio

preparation.